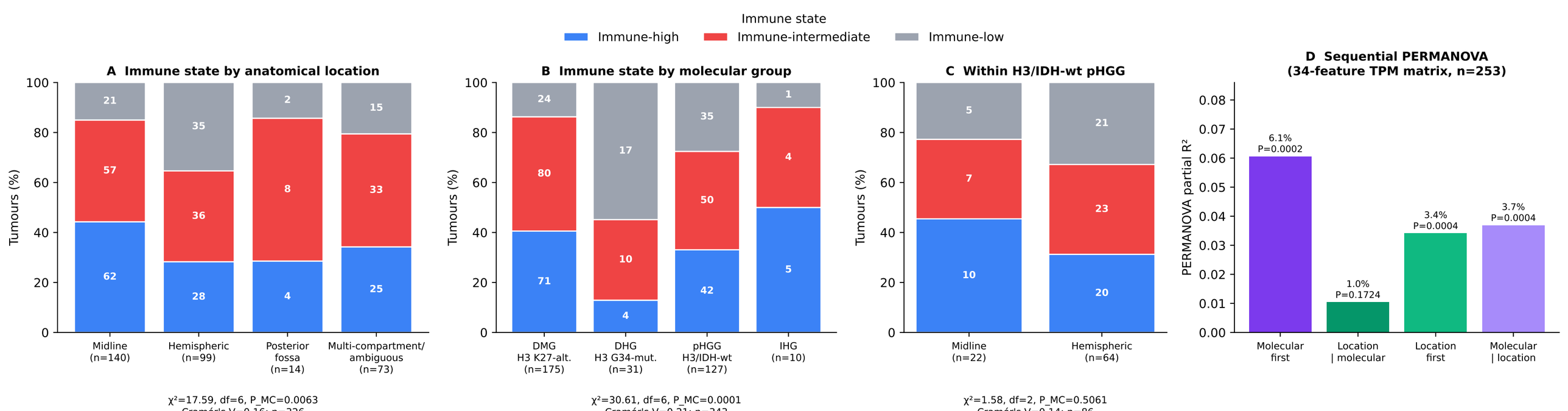

Immune state
Immune-high
Immune-intermediate
Immune-low
A Immune state by anatomical location
Tumours (%)
Midline (n=140)
Hemispheric (n=99)
Posterior fossa (n=14)
Multi-compartment/ ambiguous (n=73)
χ²=17.59, df=6, P_MC=0.0063
Cramér's V=0.16; n=326
B Immune state by molecular group
DMG H3 K27-alt. (n=175)
DHG H3 G34-mut. (n=31)
pHGG H3/IDH-wt (n=127)
IHG (n=10)
χ²=30.61, df=6, P_MC=0.0001
Cramér's V=0.21; n=343
C Within H3/IDH-wt pHGG
Midline (n=22)
Hemispheric (n=64)
χ²=1.58, df=2, P_MC=0.5061
Cramér's V=0.14; n=86
D Sequential PERMANOVA
(34-feature TPM matrix, n=253)
PERMANOVA partial R²
6.1% P=0.0002
1.0% P=0.1724
3.4% P=0.0004
3.7% P=0.0004
Molecular first
Location | molecular
Location first
Molecular | location